

CIFAR-10 full replacement CIA results

IN-replace / OUT-replace non-IID FedAvg seeds 42, 43, 44 20 rounds 5 local epochs

Recorded coverage

Active clients	3	8	16
Variants per client count	Vanilla + 3 Global-DP + 3 Metric-privacy	Vanilla + 3 Global-DP + 3 Metric-privacy	Vanilla + 3 Global-DP + 3 Metric-privacy
Noise ratios (DP)	0.002500, 0.003333, 0.006250	0.002500, 0.003333, 0.006250	0.002500, 0.003333, 0.006250
Completed IN/OUT trajectories	42 / 42	42 / 42	42 / 42

3 active clients

Table 1: Pooled 60 IN versus 60 OUT raw-loss ROC AUC; utility is the mean over rounds 16–20 and seeds 42–44.

Approach	Ratio	Noise multiplier	Clean AUC	Noisy AUC	Acc. IN	Acc. OUT	F1 IN	F1 OUT
Vanilla	–		0.957	0.851	0.800	0.808	0.800	0.808
Global-DP	0.002500	0.007500	0.882	0.759	0.800	0.808	0.799	0.808
Metric-privacy	0.002500	0.007500	0.884	0.761	0.809	0.813	0.809	0.812
Global-DP	0.003333	0.009999	0.886	0.703	0.799	0.798	0.799	0.798
Metric-privacy	0.003333	0.009999	0.882	0.742	0.803	0.811	0.803	0.811
Global-DP	0.006250	0.018750	0.892	0.559	0.734	0.736	0.731	0.735
Metric-privacy	0.006250	0.018750	0.880	0.646	0.794	0.801	0.794	0.801

8 active clients

Table 2: Pooled 60 IN versus 60 OUT raw-loss ROC AUC; utility is the mean over rounds 16–20 and seeds 42–44.

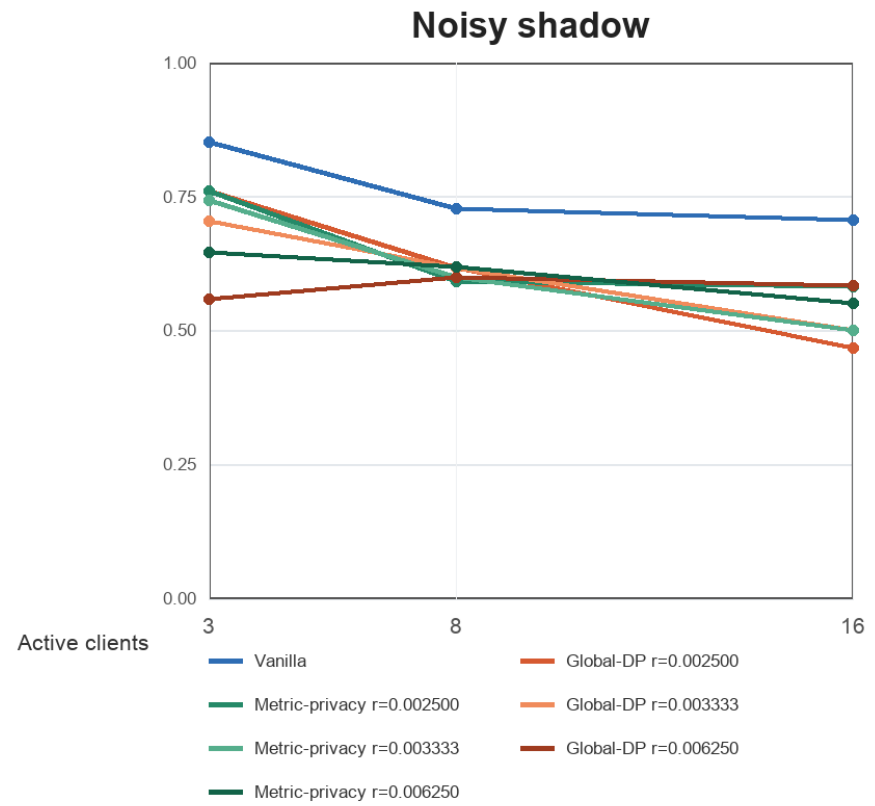
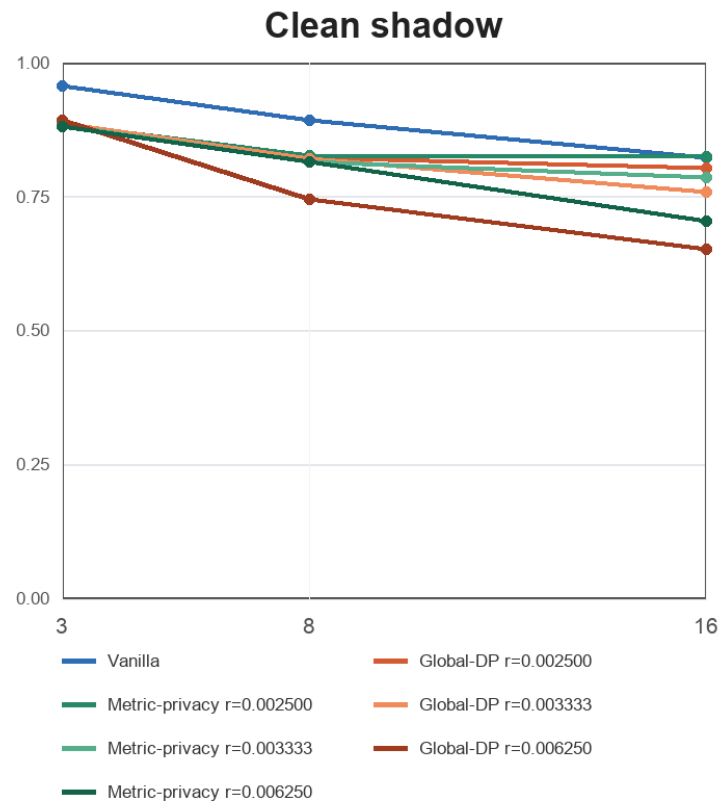
Approach	Ratio	Noise multiplier	Clean AUC	Noisy AUC	Acc. IN	Acc. OUT	F1 IN	F1 OUT
Vanilla	–		0.892	0.728	0.822	0.815	0.822	0.815
Global-DP	0.002500	0.020000	0.822	0.616	0.810	0.808	0.809	0.808
Metric-privacy	0.002500	0.020000	0.827	0.592	0.814	0.818	0.814	0.818
Global-DP	0.003333	0.026664	0.820	0.616	0.796	0.795	0.796	0.795
Metric-privacy	0.003333	0.026664	0.815	0.598	0.808	0.811	0.807	0.811
Global-DP	0.006250	0.050000	0.745	0.600	0.726	0.723	0.726	0.720
Metric-privacy	0.006250	0.050000	0.814	0.618	0.776	0.781	0.776	0.780

16 active clients

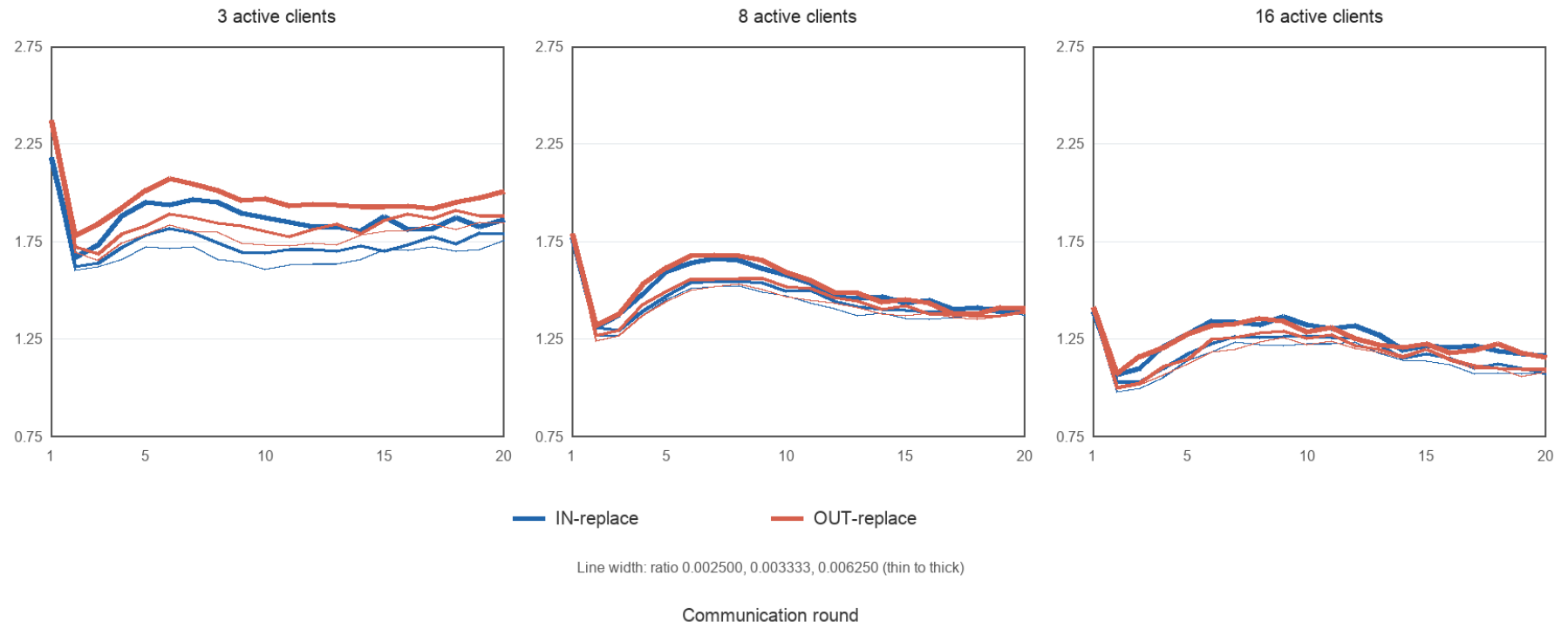
Table 3: Pooled 60 IN versus 60 OUT raw-loss ROC AUC; utility is the mean over rounds 16–20 and seeds 42–44.

Approach	Ratio	Noise multiplier	Clean AUC	Noisy AUC	Acc. IN	Acc. OUT	F1 IN	F1 OUT
Vanilla	–		0.823	0.706	0.821	0.819	0.821	0.819
Global-DP	0.002500	0.040000	0.804	0.467	0.801	0.804	0.801	0.804
Metric-privacy	0.002500	0.040000	0.823	0.582	0.805	0.807	0.805	0.807
Global-DP	0.003333	0.053328	0.759	0.501	0.788	0.790	0.788	0.790
Metric-privacy	0.003333	0.053328	0.786	0.500	0.797	0.799	0.797	0.799
Global-DP	0.006250	0.100000	0.652	0.583	0.711	0.702	0.707	0.697
Metric-privacy	0.006250	0.100000	0.705	0.550	0.743	0.741	0.742	0.738

Pooled ROC AUC across active-client counts



Metric-privacy maximum pairwise client-model distance



Source: `results/planned_runs/cifar/full/`